

RESEARCH ARTICLE

Semi-analytical solution for ultimate bearing capacity of smooth and rough circular foundations on rock considering three-dimensional strength

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Abstract

This paper proposes a semi-analytical solution for the ultimate bearing capacity q_u of both smooth and rough circular shallow foundations on rock mass. Specifically, a three-dimensional (3D) Hoek–Brown (HB) is adopted, in conjunction with equilibrium equations under axisymmetric conditions, to derive the governing equations. The method of characteristics is utilized to solve the stress and failure characteristics mesh to determine the q_u . The proposed solution is verified by using it to analyze test foundations. Comparison with an HB criterion-based solution is performed to highlight the importance of 3D strength. Furthermore, parametric studies are performed to investigate the effects of rock mass properties (intact rock constant m_i , geological strength index GSI, intact rock unconfined compressive strength σ_c) and foundation diameter (B) on the q_u , failure surface size, and vertical stress distribution on the foundation base. The results indicate that ignoring the 3D strength and the rock mass weight would lead to underestimation of q_u . Besides, the ultimate bearing capacity factor N_σ (ratio of q_u to σ_c) increases with m_i , GSI and B but decreases with σ_c . The failure surface size is significantly affected by m_i , GSI, B , σ_c and rock mass unit weight. The stress distribution on the foundation base has higher variance (higher possibility of stress concentration) at smaller σ_c , GSI, and larger B , rock mass unit weight.

KEYWORDS

base condition, characteristics method, circular foundation, rock mass, strength dependence on three principal stresses, ultimate bearing capacity

1 | INTRODUCTION

Shallow foundations are an economical foundation system to support upper structures when rock mass is at or near the ground surface. To ensure the safety of a foundation on rock mass, it is important to determine its ultimate bearing capacity (q_u) based on rock mass properties and foundation dimensions. Based on laboratory and/or field tests, different empirical relations have been proposed for estimating the q_u of shallow foundations.^{1,2} However, those empirical relations were developed based on specific test results and most of them ignore the effect of foundation dimensions. Consequently,

analytical and semi-analytical methods, which can consider both rock mass properties and foundation dimensions, have been developed for determining the q_u of shallow foundations.³

Currently, the Mohr–Coulomb (MC) criterion is often adopted for determining the q_u of shallow foundations on soil and rock mass. However, it is widely recognized that the strength of rock mass exhibits non-linearity and should be described by a non-linear criterion, for example, the Hoek–Brown (HB) criterion.⁴ So far, few approaches have adopted the HB criterion to determine the q_u of shallow foundations on rock mass. For instance, Serrano and Olalla⁵ proposed a method for determining the q_u of strip foundations on weightless rock mass based on the original HB criterion⁶ and the method of characteristics. The same approach has been extended later to anisotropic rock mass by Serrano and Olalla⁷ and to the modified MC criterion by Galindo et al.⁸ Yang et al.⁹ applied the lower bound approach to determine the q_u of a strip foundation on an HB criterion-governed rock mass. Merifield et al.¹⁰ conducted finite element limit analysis to estimate the q_u of shallow foundation on rock mass based on the HB criterion. Clausen¹¹ used the displacement finite element method to analyze the q_u of circular shallow foundations with the generalized HB criterion.⁴ Chakraborty and Kumar¹² conducted an axisymmetric quasi-lower bound finite element limit analysis to investigate the q_u of a circular foundation on rock mass following the HB criterion. Keshavarz and Kumar¹³ also used the method of characteristics to determine the q_u of both circular and strip foundations on an HB criterion-based rock mass. It is noted that although the rock mass under a shallow foundation is under a true triaxial stress condition, where all three principal stresses are different, most current studies simply ignore the intermediate principal stress and the three-dimensional (3D) strength (i.e., the strength dependence on three principal stresses) of rock mass. In this study, “3D strength” and “3D criterion” mean that all the three principal stresses are considered in the criterion and analysis, while “two-dimensional (2D) strength” and “2D criterion” represent that only the major and minor principal stresses are considered. In this regard, a “3D criterion” considers the strength dependence on three principal stresses, while a “2D criterion” considers the strength dependence only on the major and minor principal stresses. Both 3D and 2D criteria can be applied to 3D problems involving a true triaxial stress state, although a 3D criterion should be more suitable.

On the other hand, some commercial software such as FLAC3D¹⁴ and Optum G2¹⁵ with the HB criterion embedded can be used to determine the bearing capacity of a shallow foundation on rock mass, but other parameters of rock mass, such as Young’s modulus, and a user-defined or auto-generated mesh of a large zone around the foundation are needed. It can be time-consuming to use those commercial software for the design of a shallow foundation on rock mass. In this regard, to be time-efficient and consider the 3D strength of rock mass, it is necessary to develop analytical or semi-analytical solutions for the analysis of shallow foundations on rock mass.

In this study, a novel semi-analytical solution based on a 3D version of the HB criterion, the newly modified GZZ criterion,^{16,17} is proposed for determining the q_u of circular shallow foundations on rock mass. To do that, the equilibrium equations under the axisymmetric condition are considered in conjunction with the newly modified GZZ criterion, and the method of characteristics is used to derive the governing equations of the rock mass below the foundation base. A finite difference-based method is utilized to solve the governing equations so as to determine the failure surface and stresses below the circular shallow foundation with both smooth and rough base conditions. Then, the q_u of the circular shallow foundation under both smooth and rough base conditions is obtained by integrating the vertical stress on the foundation base. Verification of the proposed solution is made by comparing it with test foundations and the effect of 3D strength is investigated by comparing the proposed solution with an HB criterion-based solution.¹³ Finally, extensive parametric studies are conducted with the present solution to evaluate the effects of rock mass properties and foundation size.

2 | PROBLEM STATEMENT AND BASIC ASSUMPTIONS

2.1 | Problem statement

Figure 1 shows the failure surfaces and characteristics meshes under circular shallow foundations with smooth and rough base conditions, respectively. A centric vertical load is applied on the top of the circular shallow foundation and only half of the foundation is shown in the figure for simplicity. And the rock mass below the shallow foundation is under the axisymmetric loading conditions and thus the stresses only change within the r - z plane and the circumferential normal stress σ_θ does not change with θ and the out-of-plane shear stresses $\tau_{r\theta}$ and $\tau_{\theta z}$ are zero. Since a shallow foundation has a limited embedded depth, the effect of rock and/or soil above the foundation base is only considered as a uniformly distributed vertical stress on AD. For a smooth base foundation, the failure surface starts at the center of the foundation base and emanates to the ground surface with every point on the base failing. In contrast, for a rough base, only part of

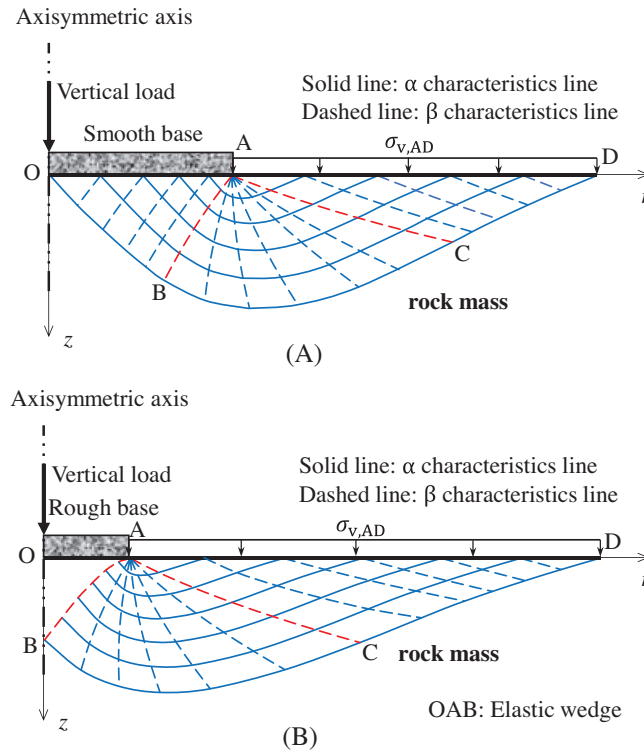


FIGURE 1 Failure surface and characteristics mesh below a circular shallow foundation: (A) smooth base; and (B) rough base.

the base approaches the failure state according to Meyerhof¹⁸ and Martin.¹⁹ As often assumed by previous studies such as Meyerhof¹⁸ and Keshavarz and Kumar,¹³ the failure occurs at the edge of a rough foundation base, and thus the innermost β-characteristic line AB is tangent to the foundation base at the edge and curvedly approaches the axisymmetric axis as shown in Figure 1B. It should be noted that the wedge zone OAB under a rough base does not fail when the failure surface is formed. And thus, the failure surface below a rough base emanates from B through C to point D under an ultimate load on OA. Especially, zone ABC is denoted as the fan zone because the curvature of this zone is larger than the other two zones. It should be noted that the shape, size, and type of the failure surfaces also depend on the material model. The present study aims to find an easy and novel method for determining the bearing capacity of a foundation on rock mass by considering the strength dependence on three principal stresses, that is, 3D strength.

2.2 | Basic assumptions

To solve the characteristics mesh and determine the q_u of a circular shallow foundation with the two base conditions, the following assumptions are made:

- 1) The rock mass below the shallow foundation is assumed to be elastic-perfectly-plastic and all rock mass within the failure surface fails at the same time when the failure surface forms. The method of characteristics is used to analyze the rock mass at failure when the elastic deformation is almost negligible compared with the plastic deformation.
- 2) Due to the true triaxial stress state of rock mass below a shallow circular foundation, the failure of the rock mass is described by a 3D version of the HB criterion, that is, the newly modified GZZ (NMGZZ) criterion, which is defined as^{16,17}

$$f_{\text{NMGZZ}} = \left[\left(\frac{q}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q}{\sigma_c} \right)^{1/a-1} \right] (I_1^* I_2^* - 9 I_3^*) - 2 m_b^2 I_3^* = 0 \quad (1)$$

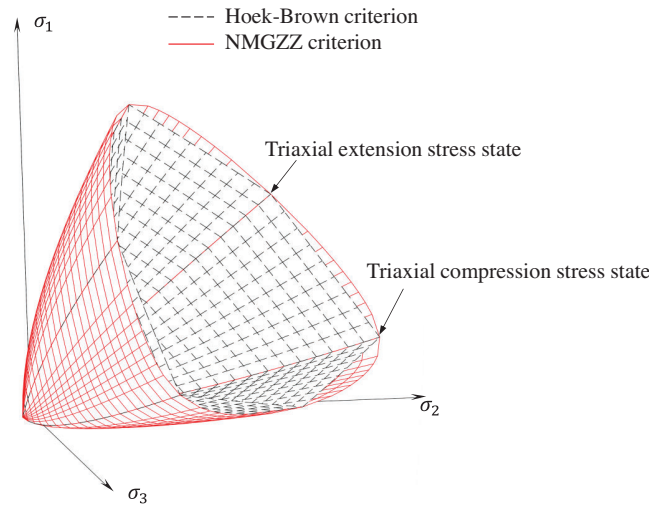


FIGURE 2 Comparison between NMGZZ criterion and HB criterion in 3D principal stress space.

where

$$q = \sqrt{I_1^{*2} - 3I_2^*} \quad (2a)$$

$$I_2^* = \sigma_1^* \sigma_2^* + \sigma_2^* \sigma_3^* + \sigma_3^* \sigma_1^* \quad (2b)$$

$$I_1^* = 3p^* = \sigma_1^* + \sigma_2^* + \sigma_3^* \quad (2c)$$

$$I_3^* = \sigma_1^* \sigma_2^* \sigma_3^* \quad (2d)$$

$$\sigma_i^* = \sigma_i + \frac{\sigma_c}{m_b} s, i = 1, 2, 3 \quad (2e)$$

$$m_b = \exp\left(\frac{GSI - 100}{28 - 14D}\right) m_i \quad (2f)$$

$$s = \exp\left(\frac{GSI - 100}{9 - 3D}\right) \quad (2g)$$

$$a = \frac{1}{2} + \frac{1}{6} \left(e^{-\frac{GSI}{15}} - e^{-\frac{20}{3}} \right) \quad (2h)$$

where σ_1 , σ_2 , and σ_3 = effective major, intermediate, and minor principal stresses, respectively; m_i = a constant of intact rock; m_b , a and s = parameters of rock mass; GSI = geology strength index and is proposed by Hoek²⁰ for the estimation of rock mass strength and rock mass deformation modulus; and D = a disturbance factor representing the effect of construction on rock mass.

The comparison between the NMGZZ criterion and the HB criterion in the 3D principal stress space is shown in Figure 2. As can be seen, the NMGZZ criterion perfectly coincides with the HB criterion under the triaxial compression and extension stress states and thus it can be regarded as a 3D revision of the HB criterion for modeling the behavior of rock mass and performs better than the original HB criterion under true triaxial stress states. More details about the NMGZZ criterion can be found in Chen et al.^{16,17}

- 3) According to Drescher and Detournay,²¹ for the bearing capacity problem, the non-associated flow rule would lead to a velocity field violating the thermodynamic requirement of non-negative energy dissipation and thus the associated flow rule is assumed for the rock mass. The elastic deformation before failure is much smaller than the plastic deformation at failure. In this regard, the plastic strain rate is in the direction of the movement (velocity) of the rock mass at failure, and the velocity field (characteristics mesh) coincides with the stress field.

- 4) The lateral strain of the ground is assumed to be zero before the installation of the foundation, which is referred to as the initial condition thereafter. Therefore, the coefficient of lateral pressure at rest K_0 is related to the Poisson's ratio of the rock mass ν_r as

$$K_0 = \frac{\sigma_{h0}}{\sigma_{v0}} = \frac{\nu_r}{1 - \nu_r} \quad (3a)$$

where σ_{h0} , σ_{v0} = initial lateral and vertical effective stresses, respectively; K_0 = coefficient of lateral pressure at rest; and ν_r = Poisson's ratio of rock mass.

Alternatively, the initial stresses can be related to each other as follows:

$$\sigma_{h0} = K_0 \sigma_{v0} = \nu_r (\sigma_{v0} + \sigma_{h0}) \quad (3b)$$

- 5) For a circular foundation, the effective circumferential stress keeps changing during loading and can be expressed as

$$\begin{aligned} \sigma_\theta &= E_{rm} \epsilon_\theta^e + \sigma_{\theta0} + \nu_r (\sigma_r + \sigma_z) - \nu_r (\sigma_{r0} + \sigma_{z0}) \\ &= E_{rm} \epsilon_\theta^e + \sigma_{h0} + \nu_r (\sigma_r + \sigma_z) - \nu_r (\sigma_{h0} + \sigma_{v0}) \end{aligned} \quad (4a)$$

$$\epsilon_\theta^e = -\frac{u_r}{r} \quad (4b)$$

where σ_r , σ_θ , and σ_z = effective stresses in the radial, circumferential and vertical directions under a circular foundation, respectively; E_{rm} = Young's modulus of the rock mass; σ_{r0} , $\sigma_{\theta0}$ and σ_{z0} = initial values of σ_r , σ_θ and σ_z , respectively; ϵ_θ^e = elastic circumferential strain; u_r = radial displacement; and r = radial coordinate. Note that Equation (4) applies to rock mass along the failure surface because the rock mass is perfectly plastic and all rock mass fails at the same time.

During loading, both the σ_r and σ_z increase due to compression from the foundation base. According to Equation (4), the σ_θ also increases and its value at failure would be higher than the earth lateral pressure at rest σ_{h0} . Therefore, considering a larger σ_θ than σ_{h0} leads to higher σ_z on the foundation base and thus the increase of the bearing capacity compared to the result with $\sigma_\theta = \sigma_{h0}$. For conservative consideration, ϵ_θ^e is taken as zero and thus Equation (4a), by considering Equation (3b), reduces to

$$\sigma_\theta = \nu_r (\sigma_r + \sigma_z) \quad (5)$$

In this regard, the initial stress state would not affect the relationship among σ_θ , σ_r and σ_z because the initial stress components are eliminated from Equation (5).

- 6) Since rock mass is under the axisymmetric condition, and $\tau_{r\theta}$ and $\tau_{\theta z}$ are zero, the circumferential normal stress σ_θ is a principal stress. And the failure takes place within the r - z plane, which means σ_1 and σ_3 are within the r - z plane, thus $\sigma_2 = \sigma_\theta$.
- 7) For a smooth base, the friction angle of the interface between rock mass and foundation base ϕ is zero, and thus the effective major principal stress on the foundation base is in the vertical direction.
- 8) For a rough base, a non-plastic wedge, as shown in Figure 1B, is assumed to occur below the foundation base according to Meyerhof.¹⁸ The details about the non-plastic wedge are discussed in section 3.2.

3 | GOVERNING EQUATIONS AND BOUNDARY CONDITIONS

3.1 | Governing equations

The equilibrium equations along the radial and vertical directions considering the axisymmetric conditions are

$$\frac{\partial \sigma_r}{\partial r} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_r - \sigma_\theta}{r} = 0 \quad (6a)$$

$$\frac{\partial \tau_{rz}}{\partial r} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{rz}}{r} = \gamma_{rm} \quad (6b)$$

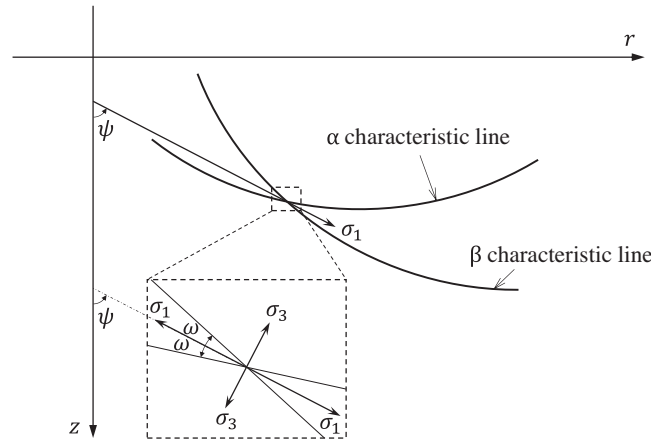


FIGURE 3 Relationships between characteristic lines and direction of effective major principal stress σ_1 .

where τ_{rz} = shear stress along the radial and vertical directions; and γ_{rm} = unit weight of rock mass. Note that the equilibrium equation in the circumferential direction is not shown in Equation (6) because σ_θ is constant with respect to θ and shear stresses $\tau_{r\theta}$ and $\tau_{\theta z}$ are zero under axisymmetric conditions and thus the equilibrium equation in the circumferential direction is satisfied.

Considering the failure criterion in Equation (1) and the relation between different stress components in Equation (5) under the axisymmetric condition, the newly modified GZZ criterion under the axisymmetric condition is further simplified and expressed by two Lambe parameters p_L and q_L as²²

$$f_{\text{NMGZZ}} = F(p_L, q_L) \quad (7)$$

where

$$p_L = (\sigma_1 + \sigma_3) / 2 \quad (8a)$$

$$q_L = (\sigma_1 - \sigma_3) / 2 \quad (8b)$$

Based on the direction of the effective major principal stress ψ and with $\sigma_\theta = \sigma_2$, the stresses can be further simplified as

$$\sigma_r = p_L - q_L \cos 2\psi \quad (9a)$$

$$\sigma_z = p_L + q_L \cos 2\psi \quad (9b)$$

$$\tau_{rz} = q_L \sin 2\psi \quad (9c)$$

$$\sigma_\theta = 2\nu_r p_L \quad (9d)$$

where ψ = the counterclockwise angle from the positive z -direction to the direction of the effective major principal stress as shown in Figure 3.

The differential equations of the two groups of characteristic lines in the failure mesh are:

$$\frac{dr}{dz} = \tan(\psi \pm \omega) \quad (10a)$$

$$\omega = \frac{\pi}{4} - \frac{\rho}{2} \quad (10b)$$

where Equation (10a) with “+” and “−” sign governs the α and β characteristic lines, respectively; ρ = instant friction angle of rock mass; and ω is self-evident. Following Chen et al.²²

$$\sin \rho = \frac{dq_L}{dp_L} = \frac{d\sigma_1 - d\sigma_3}{d\sigma_1 + d\sigma_3} = \frac{1 - C_{13}}{C_{13} + 1} \quad (11a)$$

$$C_{13} = -\frac{\frac{\partial f}{\partial \sigma_1^*} + \nu_r \frac{\partial f}{\partial \sigma_2^*}}{\frac{\partial f}{\partial \sigma_3^*} + \nu_r \frac{\partial f}{\partial \sigma_2^*}} \quad (11b)$$

Substituting Equations (6) and (9) into Equation (10):

$$\begin{aligned} \cos \rho dp_L + 2q_L d\psi + \frac{(p_L - \sigma_\theta) \cos \rho - q_L \cos(2\psi - \rho)}{r} dr \\ + \frac{(p_L - \sigma_\theta) \sin \rho + q_L \sin(2\psi - \rho)}{r} dz = \gamma \cos \rho dz - \gamma \sin \rho dr \end{aligned} \quad (12a)$$

$$\begin{aligned} \cos \rho dp_L - 2q_L d\psi + \frac{(p_L - \sigma_\theta) \cos \rho - q_L \cos(2\psi + \rho)}{r} dr \\ - \frac{(p_L - \sigma_\theta) \sin \rho - q_L \sin(2\psi + \rho)}{r} dz = \gamma \cos \rho dz + \gamma \sin \rho dr \end{aligned} \quad (12b)$$

where Equations (12a) and (12b) are the stress equilibrium differential equations along the α and β characteristic lines, respectively. Equations (10) to (12) together constitute the governing equations of the failure characteristics mesh.

3.2 | Boundary conditions

To solve the governing Equations (10) to (12) so as to determine the stresses on OA, the stress state, the direction of the effective major principal stress ψ , and the coordinates of points on AD as well as ψ and coordinates of points on OA should be known. This section discusses the boundary conditions of AD and OA under both smooth and rough base conditions.

3.2.1 | Boundary AD

To be conservative, the effect of rock and/or soil above the foundation base is only considered as a uniformly distributed vertical stress on AD. It is assumed that the vertical stress condition on AD is not affected by the load on the foundation base and the base condition. So, the effective major principal stress should be in the horizontal direction. Therefore,

$$\psi_{AD} = 90^\circ \quad (13a)$$

$$\sigma_{3,AD} = \sigma_{v,AD} = \gamma_{rm} H_{r1} \quad (13b)$$

where H_{r1} = depth of rock mass above the foundation base, with H_{r1} being 0 for a shallow foundation on the ground surface.

Considering the axisymmetric condition Equations (5) and (13b), the failure criterion Equation (1) can be simplified as a polynomial equation, and it can be numerically solved to determine $\sigma_{1,AD}$ and $\sigma_{2,AD}$.

3.2.2 | Boundary OA

1) Smooth base

For a smooth foundation, no shear stress occurs on the base and thus the effective major principal stress should be in the vertical direction. Hence,

$$\psi_{OA} = 0^\circ \quad (14)$$

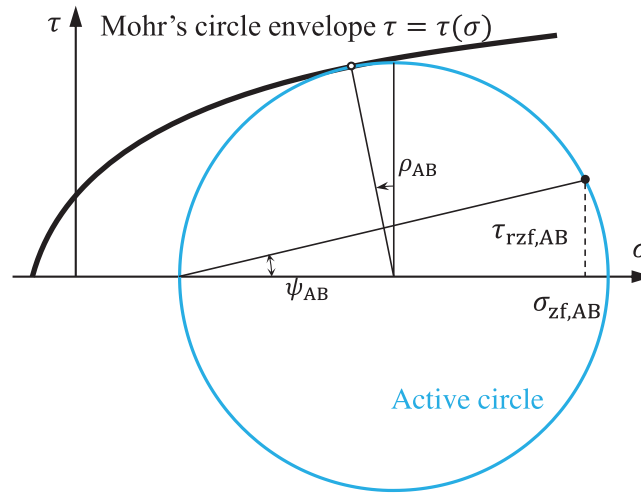


FIGURE 4 Mohr circle for points on AB with a rough base condition.

2) Rough base

According to Meyerhof¹⁸ and Frydman and Burd,²³ the failure of rock mass partially takes place on the interface of the foundation and rock mass close to the foundation edge for a perfectly rough foundation. This condition can be satisfied when a β characteristics line is tangential to the foundation base at the edge of the foundation (point A) as shown in Figure 1B. The stress components at point B should be symmetric and thus $\sigma_{1,B}$ is in the vertical direction. Under this context, no failure takes place within the non-plastic zone OAB, and thus the boundary conditions along AB should alternatively be determined.

Note that the shear stress on the base increases from 0 at point O to the maximum at point A. Since the rock mass fails at point A, the friction angle of the rock mass-foundation interface at point A ϕ_A is simply ρ_A , where ρ_A is the friction angle of the rock mass close to point A. Therefore, at point A,

$$\psi_A - \omega_A = -90^\circ \quad (15a)$$

or

$$\psi_A = -45^\circ - \frac{\rho_A}{2} \quad (15b)$$

As for point B,

$$\psi_B = 0^\circ \quad (16)$$

According to Equation (9) and Figure 4, the vertical normal and horizontal shear stresses for points on AB at failure can be related to the principal stresses as

$$\sigma_{zf,AB} = \frac{\sigma_{1f,AB} + \sigma_{3f,AB}}{2} + \frac{\sigma_{1f,AB} - \sigma_{3f,AB}}{2} \cos 2\psi_{AB} = p_{Lf,AB} + q_{Lf,AB} \cos 2\psi_{AB} \quad (17a)$$

$$\tau_{rzf,AB} = \frac{\sigma_{1f,AB} - \sigma_{3f,AB}}{2} \sin 2\psi_{AB} = q_{Lf,AB} \sin 2\psi_{AB} \quad (17b)$$

where $\sigma_{zf,AB}$, $\tau_{rzf,AB}$ = effective normal and shear stresses along the vertical and horizontal directions for points on AB, respectively; $\sigma_{1f,AB}$ and $\sigma_{3f,AB}$ = effective major and minor principal stresses for points on AB at failure, respectively. And all those stresses $\sigma_{zf,AB}$, $\tau_{rzf,AB}$, $\sigma_{1f,AB}$ and $\sigma_{3f,AB}$ and ψ_{AB} change with locations.

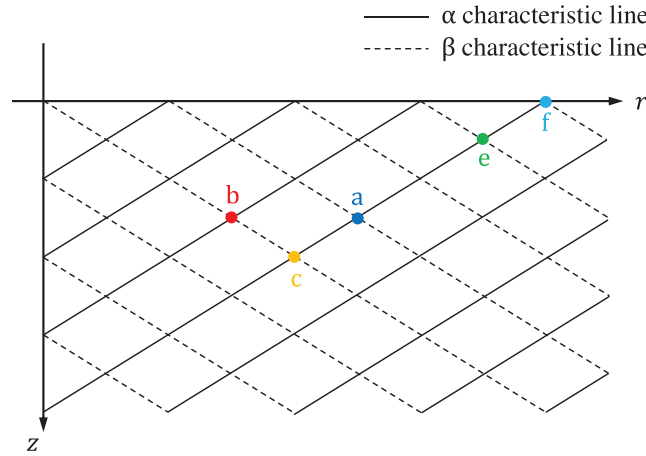


FIGURE 5 Points on characteristics mesh.

4 | SOLUTION

4.1 | Solution to stresses within failure surface (characteristics mesh)

A finite difference-based procedure is proposed to determine the stresses, coordinates (r, z) , and ψ for each point on the characteristics mesh by considering Equations (10) to (12) and boundary conditions given by Equations (13) to (17). As shown in Figure 1, for each point on the characteristics mesh below OA and AD, there are two adjacent points along the α and β characteristic lines, respectively. Therefore, each point below plane OAD can be regarded as point c in Figure 5, and the two adjacent points a and b with determined stresses can be easily located within the characteristics mesh. Substituting the stresses, coordinates, and ψ of points a, b and c into Equations (10) to (12) and using the central finite difference formulas of the governing equations, one can obtain

$$r_c = \frac{r_b \tan(\psi_{ac} + \omega_{ac}) - r_a \tan(\psi_{bc} - \omega_{bc}) + (z_a - z_b) \tan(\psi_{ac} + \omega_{ac}) \tan(\psi_{bc} - \omega_{bc})}{\tan(\psi_{ac} + \omega_{ac}) - \tan(\psi_{bc} - \omega_{bc})} \quad (18a)$$

$$z_c = \frac{r_b - r_a + z_a \tan(\psi_{ac} + \omega_{ac}) - z_b \tan(\psi_{bc} - \omega_{bc})}{\tan(\psi_{ac} + \omega_{ac}) - \tan(\psi_{bc} - \omega_{bc})} \quad (18b)$$

$$\psi_c = \frac{\Delta_{ac} / \cos \rho_{ac} - \Delta_{bc} / \cos \rho_{bc} + 2(\psi_a q_{Lac} / \cos \rho_{ac} + \psi_b q_{Lbc} / \cos \rho_{bc}) + (p_{La} - p_{Lb})}{2(q_{Lac} / \cos \rho_{ac} + q_{Lbc} / \cos \rho_{bc})} \quad (18c)$$

$$p_{Lc} = \frac{\Delta_{ac} q_{Lbc} + \Delta_{bc} q_{Lac} + 2q_{Lac} q_{Lbc} (\psi_a - \psi_b) + p_{La} q_{Lbc} \cos \rho_{ac} + p_{Lb} q_{Lac} \cos \rho_{bc}}{q_{Lac} \cos \rho_{bc} + q_{Lbc} \cos \rho_{ac}} \quad (18d)$$

where the items with subscripts “a”, “b”, and “c” represent the stresses, coordinates and ψ for points a, b, and c, respectively; the items with subscripts “ac” and “bc” are the average of points a and c and the average of points b and c, respectively; and Δ_{ac} and Δ_{bc} are

$$\Delta_{ac} = - \left[\gamma_{rm} \sin \rho_{ac} + \frac{(p_{Lac} - \sigma_{\theta ac}) \cos \rho_{ac} - q_{Lac} \cos(2\psi_{ac} - \rho_{ac})}{r_{ac}} \right] (r_c - r_a) + \left[\gamma_{rm} \cos \rho_{ac} - \frac{(p_{Lac} - \sigma_{\theta ac}) \sin \rho_{ac} + q_{Lac} \sin(2\psi_{ac} - \rho_{ac})}{r_{ac}} \right] (z_c - z_a) \quad (19a)$$

$$\Delta_{bc} = \left[\gamma_{rm} \sin \rho_{bc} - \frac{(p_{Lbc} - \sigma_{\theta bc}) \cos \rho_{bc} - q_{Lbc} \cos(2\psi_{bc} + \rho_{bc})}{r_{bc}} \right] (r_c - r_b) + \left[\gamma_{rm} \cos \rho_{bc} + \frac{(p_{Lbc} - \sigma_{\theta bc}) \sin \rho_{bc} - q_{Lbc} \sin(2\psi_{bc} + \rho_{bc})}{r_{bc}} \right] (z_c - z_b) \quad (19b)$$

For a point at the edge of the foundation, it fails regardless of the base condition and is described by a degenerate α characteristic line according to Martin,¹⁹ which can serve as a “base” characteristic line for building up the whole mesh. And the degenerate α characteristic line satisfies:

$$\cos \rho dp_L + 2q_L d\psi = 0 \quad (20)$$

To solve Equation (20), the degenerate α characteristic line is divided into n divisions with each division point corresponding to a specific ψ_i . Then, each division point at ψ_i can be determined by numerically integrating Equation (20) from ψ_{AD} to ψ_i ($\psi_{AD} < \psi_i \leq \psi_{OA}$).

To solve the characteristics mesh below the foundation base, it is required to repetitively perform Equations (18) and (19) to generate the characteristics mesh. Both the stresses and ψ for each point on the characteristics mesh can be obtained when determining the mesh. To do that, first, the adjacent points on boundary AD can be taken as the first set of points a and b, and every intersection point of the mesh a little bit below boundary AD can be taken as point c and solved by using Equations (18) and (19). Note that the distance between adjacent points on AD is predefined and the smaller the more accurate. Then, by repetitively taking the newly determined points from the last step as the new set of points a and b and performing analysis using Equations (18) and (19) to solve new point c, the characteristics mesh can be gradually generated from boundary AD to points near OA (smooth base) or on AB (rough base).

However, when the characteristics mesh develops to OA, only one adjacent point along the α characteristic line and on OA can be found as points on OA might not be an intersection point of the developed characteristics mesh. In this case, the point determined from the last step is taken as point e, and its adjacent point along the α characteristic line and on OA can be taken as point f in Figure 5. Therefore, the boundary conditions on OA should be utilized as additional information for solving point f. For each point on OA, the vertical coordinate and ψ_{OA} are

$$z_{OA} = 0 \quad (21a)$$

$$\psi_{OA} = 0 \quad (21b)$$

Considering Equation (21) and the governing equations along the α characteristic line

$$r_f = r_e - z_e \tan(\psi_{ef} + \omega_{ef}) \quad (22a)$$

$$p_{Lf} = p_{Le} + \frac{\Delta_{ef} + 2q_{Lef}\psi_{ef}}{\cos \rho_{ef}} \quad (22b)$$

in which,

$$\begin{aligned} \Delta_{ef} = & - \left[\gamma_{rm} \sin \rho_{ef} + \frac{(p_{Lef} - \sigma_{\theta ef}) \cos \rho_{ef} - q_{Lef} \cos(2\psi_{ef} - \rho_{ef})}{r_{ef}} \right] (r_f - r_e) \\ & - \left[\gamma_{rm} \cos \rho_{ef} - \frac{(p_{Lef} - \sigma_{\theta ef}) \sin \rho_{ef} + q_{Lef} \sin(2\psi_{ef} - \rho_{ef})}{r_{ef}} \right] z_e \end{aligned} \quad (23)$$

where the items with subscript “ef” denote the average of points e and f.

Finally, with the help of Equation (23), the characteristics mesh is successfully extended to the foundation base and thus the stresses within the failure surface are determined.

4.2 | Solution to ultimate bearing capacity

By utilizing the proposed finite difference-based procedures, the stresses within the failure surface are obtained. Based on the force equilibrium of the foundation in the vertical direction, the q_u of a shallow circular foundation under smooth and rough base conditions can be determined by the integration of the vertical stress on the foundation base as

$$q_u = \frac{8}{B^2} \int_0^{B/2} \sigma_{zf,OA}(r) r dr, \text{ smooth base} \quad (24a)$$

TABLE 1 Rock mass properties, foundation diameter, and ultimate bearing capacity of test foundations.

Shaft no.	Rock type	m_i	σ_c (MPa)	RMR	GSI	γ_{rm} (kN/m ³)	B (m)	q_{um} (MPa)	Measured $N_{\sigma m}$	Predicted $N_{\sigma p}$		Source
										Smooth	Rough	
1	Chalk	7	0.31	15	10	19.25	0.87	0.30	0.97	0.22	0.89	Lord ²⁶
2	Marlstone	5	13.80	62	57	27.67	0.15	20.00	1.45	1.33	1.45	Maleki and Hollberg ²⁷
3	Tuff	13	4.03	70	65	^a	0.30	14.65	3.64	3.64	4.25	Pellegrino ²⁸
4	Tuff	13	4.03	70	65	^a	0.30	16.00	3.97	3.65	4.25	Pellegrino ²⁸
5	Tuff	13	3.35	75	70	^a	0.30	12.00	3.58	3.37	5.20	Pellegrino ²⁸
6	Tuff	13	2.00	65	60	^a	0.30	5.92	2.96	3.13	3.68	Pellegrino ²⁸
7	Sandstone	17	102.97	90	85	^a	0.30	1029.78	10.00	10.39	11.46	Pells and Turner ^{29,30}
8	Sandstone	17	44.99	90	85	^a	0.20	404.93	9.00	10.36	11.26	Pells and Turner ^{29,30}
9	Sandstone	17	74.96	90	85	^a	0.20	674.82	9.00	10.37	11.25	Pells and Turner ^{29,30}
10	Limestone	10	124.00	80	85	25	0.02	558.38	4.50	4.63	4.94	Rehman and Broms ³¹
11	Chalk	7	0.90	15	10	^a	0.14	0.48	0.53	0.11	0.63	Burland and Lord ³²
12	Chalk	7	0.82	55	50	^a	0.14	1.00	1.22	1.29	2.00	Burland and Lord ³²
13	Chalk	7	1.04	87	82	^a	0.14	5.27	5.07	4.81	5.61	Lake and Simons ³³

^aRock mass unit weight γ_{rm} was not given and is taken as 22 kN/m³.

$$q_u = \frac{8}{B^2} \int_0^{B/2} \left(\sigma_{zf,AB} + \tau_{rzf,AB} \frac{\partial z}{\partial r} + \gamma_{rm} z \right) r dr, \text{ rough base} \quad (24b)$$

5 | VALIDATION AND COMPARISON

5.1 | Verification with test foundations

To verify the proposed solution, thirteen (13) test foundations were collected from different sources. Table 1 summarizes the rock mass properties, foundation diameter B , and measured ultimate bearing capacity q_{um} of the test foundations. Among those test foundations, foundations 1–9 and 11–13 are from field tests, and foundation 10 is from a laboratory test. For convenient comparison, the measured ultimate bearing capacity factor $N_{\sigma m}$, which is the ratio of q_{um} to σ_c , is also shown in the table. For all cases, the m_i was determined based on the rock type following Hoek and Brown,²⁴ σ_c and B were taken directly from the original source, and GSI was determined from the given RMR (Rock Mass Rating) proposed by Bieniawski²⁵ and using the following relation²⁴

$$GSI = RMR - 5 \quad (25)$$

Since the installation of a shallow foundation most likely has only minor effects on rock mass, a disturbance factor $D = 0$ was used. By using the rock mass properties and foundation diameters from Table 1, the predicted ultimate bearing capacity factor $N_{\sigma p}$ (the ratio of predicted q_{up} to σ_c) of each test foundation was determined with the proposed solution for both smooth and rough base conditions and the obtained predicted $N_{\sigma p}$ values are shown in Table 1 and plotted in Figure 6. As expected, the predicted $N_{\sigma p}$ with a rough base condition is larger than that with a smooth base condition. The measured $N_{\sigma m}$ falls between the predicted $N_{\sigma p}$ lines at smooth and rough base conditions or is very close to them, which verifies the proposed solution. For practical applications, it is recommended to take the solution at the smooth base condition as the lower bound and that at the rough base condition as the upper bound, and thus the solution at smooth base condition as a conservative estimation for the ultimate bearing capacity of a circular shallow foundation on rock mass.

5.2 | Comparison with 2D HB criterion-based solution

To further evaluate the proposed solution and highlight the effect of 3D strength, the proposed solution is compared with a 2D HB criterion-based solution by Keshavarz and Kumar.¹³ Both the smooth and rough base conditions are considered in the comparison.

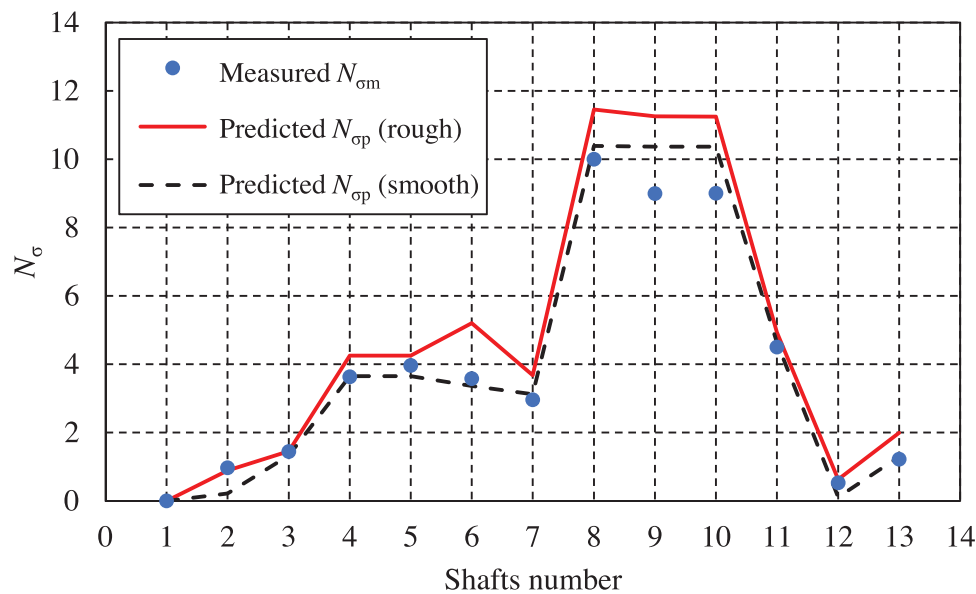


FIGURE 6 Comparisons between predicted and measured ultimate bearing capacity factor (N_σ) values.

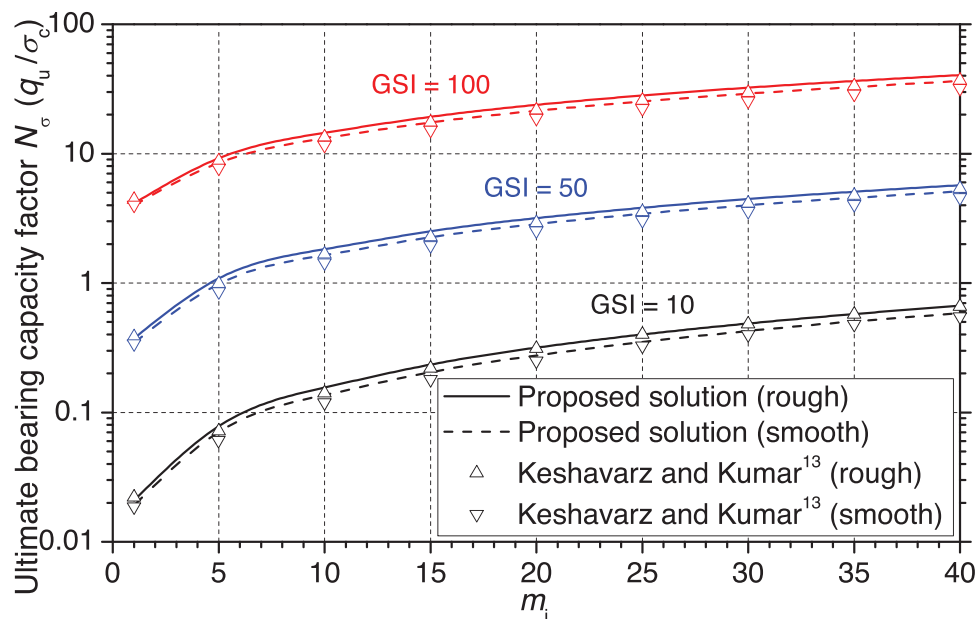


FIGURE 7 Variation of N_σ with m_i under different GSI.

Figure 7 shows a comparison between the proposed solution and the 2D HB criterion-based solution for circular foundations with different m_i , GSI = 10, 50, and 100, and under the two base conditions. $\sigma_c = 5$ MPa, $D = 0$, $\gamma_{rm} = 0$ kN/m³, and $B = 1$ m are considered based on the results available in Keshavarz and Kumar.¹³ As shown in Figure 7, for both the proposed solution and the 2D HB criterion-based solution, the N_σ increases with larger m_i and is significantly enhanced with the increase of GSI. Besides, the proposed solution predicts a higher N_σ than the 2D HB criterion-based solution at the same GSI, m_i and base condition, indicating the importance of considering the intermediate principal stress or 3D strength. Also, both solutions give a higher N_σ at a rough base condition than at a smooth base condition. To better understand the effect of 3D strength, the difference between the present solution and the solution of Keshavarz and Kumar¹³ is also shown in Table 2. As can be seen, the difference between the present solution and the previous solution¹³ can be over 10%. Therefore, it is important to consider 3D strength and base condition when determining the q_u of a circular shallow foundation on rock mass.

TABLE 2 Comparison of the present solution and the solution by Keshavarz and Kumar.¹³

m_i	Difference between present solution and solutions using 2D HB criterion $(N_{\sigma p3D} - N_{\sigma p2D})/N_{\sigma p2D}$ ^a					
	GSI = 10, rough	GSI = 10, smooth	GSI = 50, rough	GSI = 50, smooth	GSI = 100, rough	GSI = 100, smooth
1	5.61%	1.71%	2.13%	1.17%	6.74%	0.43%
5	9.01%	16.00%	9.82%	13.98%	5.38%	9.50%
10	10.52%	14.08%	9.52%	14.28%	8.41%	13.10%
15	7.12%	13.56%	10.96%	15.52%	10.71%	14.30%
20	1.63%	10.14%	8.79%	12.63%	9.65%	14.20%
25	0.00%	7.22%	7.98%	12.36%	9.38%	13.07%
30	1.45%	6.55%	7.88%	11.64%	9.59%	13.82%
35	0.66%	5.19%	7.13%	12.13%	9.41%	13.60%
40	1.67%	6.26%	6.39%	11.39%	10.07%	13.60%

^a $N_{\sigma p3D}$ = predicted $N_{\sigma p}$ of the present solution, $N_{\sigma p2D}$ = result from Keshavarz and Kumar.¹³

TABLE 3 Rock mass properties and shaft dimensions for parametric analyses.

#	m_i	σ_c (MPa)	GSI	γ_{rm} (kN/m ³)	B (m)	D
A	5	5	50	22 or 0	1	0
b	15	5	50	22 or 0	1	0

6 | PARAMETRIC STUDIES

In this section, parametric studies are performed with the proposed solution to study the effect of rock mass properties (m_i , σ_c and GSI) and foundation size (diameter) B on the foundation performance. The N_{σ} , the size of the failure surface and the vertical stress distribution on the foundation base are systematically investigated. Table 3 shows the parameters of the reference cases, and the parametric analyses are performed by changing one parameter at a time.

6.1 | Effects of σ_c

Figure 8 shows the effect of σ_c on N_{σ} with $m_i = 5$ and 15, $\gamma_{rm} = 0$ and 22 kN/m³, and both smooth and rough bases. As expected, the N_{σ} with a higher m_i is larger than that with a smaller m_i . Furthermore, the N_{σ} decreases with σ_c if $\gamma_{rm} > 0$ but keeps constant when $\gamma_{rm} = 0$. The N_{σ} with $\gamma_{rm} > 0$ is large than that at $\gamma_{rm} = 0$ and their difference decreases with a higher σ_c and is almost negligible when σ_c is greater than 30 MPa. Therefore, it is important to consider the unit weight of rock mass when determining the ultimate bearing capacity of a circular shallow foundation, especially when the rock mass has a low σ_c .

Figure 9 shows the effect of σ_c on the normalized failure surface size with $m_i = 5$ and 15, $\gamma_{rm} = 0$ and 22 kN/m³, and both smooth and rough base conditions, where the black solid lines are the failure surfaces with $\gamma_{rm} > 0$ and the red dashed line represents those with $\gamma_{rm} = 0$. The sizes of the failure surfaces are normalized by the diameter of the circular shallow foundation. As can be seen, the failure surface with $m_i = 15$ is larger than that with $m_i = 5$ at the same σ_c and base condition. Furthermore, the failure surface with $\gamma_{rm} = 0$ does not change with σ_c and is larger than those with $\gamma_{rm} > 0$ at both rough and smooth base conditions.

Figure 10 shows the effect of σ_c on the normalized vertical stress distribution on the smooth foundation base at $m_i = 5$ and 15 and $\gamma_{rm} = 0$ and 22 kN/m³. The black solid lines are the normalized vertical stress distributions with $\gamma_{rm} > 0$ and the red dashed lines represent those with $\gamma_{rm} = 0$. Similar to the N_{σ} , the normalized vertical stresses increase with larger m_i at the same σ_c . Furthermore, the normalized vertical stress decreases and the decreasing rate first decreases then increases with the increase of distance from the foundation center. The normalized vertical stress with $\gamma_{rm} = 0$ is not affected by σ_c and is smaller than those with $\gamma_{rm} > 0$ at the same σ_c . And the normalized vertical stress with $\gamma_{rm} > 0$ decreases with higher σ_c and gradually approaches that for $\gamma_{rm} = 0$. This is again because the effect of unit weight is smaller when σ_c is higher. Therefore, the variance of the normalized vertical stress on the foundation base is higher when σ_c is smaller, and stress concentration might occur for a circular shallow foundation on rock mass with a low σ_c .

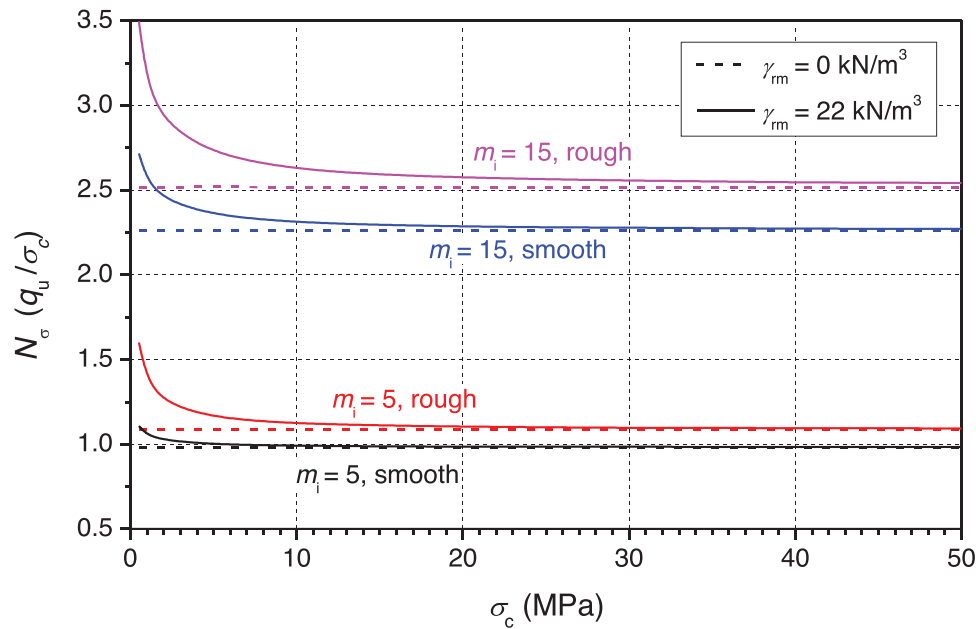


FIGURE 8 Relation between σ_c and N_σ .

Figure 11 shows the effect of σ_c on the normalized vertical stress distribution on the foundation base with a rough base condition at $m_i = 5$ and 15 and $\gamma_{rm} = 0$ and 22 kN/m³. The trend of the normalized vertical stress distribution is similar to that at a smooth base condition except that the normalized vertical stress on a rough base is always larger than that on a smooth base at the same m_i and σ_c . Furthermore, different from the smooth base case, the normalized vertical stress at the edge of the foundation decreases with higher σ_c because of the more significant effect of the rock mass weight at a lower σ_c .

6.2 | Effect of GSI

Figure 12 shows the effect of GSI on N_σ with $m_i = 5$ and 15, $\gamma_{rm} = 0$ and 22 kN/m³, and both smooth and rough base conditions. As can be seen, the N_σ increases with higher GSI at both base conditions. At the same GSI, N_σ increases with higher m_i , larger γ_{rm} and when the base condition changes from smooth to rough.

Figure 13 shows the effect of GSI on the normalized failure surface size with $m_i = 5$, $\gamma_{rm} = 0$ and 22 kN/m³, and both smooth and rough base conditions, where the solid lines are the failure surfaces with $\gamma_{rm} > 0$ and the dashed lines represent those with $\gamma_{rm} = 0$. To save space, only the cases with $m_i = 5$ are plotted in Figure 13 and similar trends can be found for the cases with $m_i = 15$. It is interesting to find that the normalized failure surface with $\gamma_{rm} = 0$ contracts with higher GSI under the two base conditions. This is because the rock mass under the base with a higher GSI has a larger friction angle, leading to a fan zone with a higher curvature and thus a smaller normalized failure surface. In contrast, the normalized failure surface with $\gamma_{rm} > 0$ first expands and then shrinks with the increase of GSI under the two base conditions. This is because the effect of γ_{rm} , which enhances the bearing capacity and decreases the failure surface size, varies with GSI. When $GSI < 30$, the effect of γ_{rm} dominates and the normalized failure surface expands with GSI increasing from 10 to 30 as the effect of γ_{rm} decreases with higher GSI or rock mass strength. After that, when $30 < GSI < 50$, the effect of γ_{rm} decreases while the effect of GSI increases, leading to a barely changed normalized failure surface. When $GSI > 50$, the effect of γ_{rm} is almost negligible because the strength of rock mass is much higher than the stress induced by the weight of rock mass, and thus the effect of GSI dominates, resulting in the shrinkage of the normalized failure surface. Again, the failure surface at a rough base condition is larger than that at a smooth base condition. Therefore, the failure surface size is influenced by GSI, γ_{rm} and base condition and the various factors should be all considered in the design of a circular shallow foundation on rock mass.

Figure 14 shows the effect of GSI on the normalized vertical stress distribution on the foundation base with $m_i = 5$, $\gamma_{rm} = 0$ and 22 kN/m³, and both smooth and rough base conditions. Again, only the cases with $m_i = 5$ is shown here and

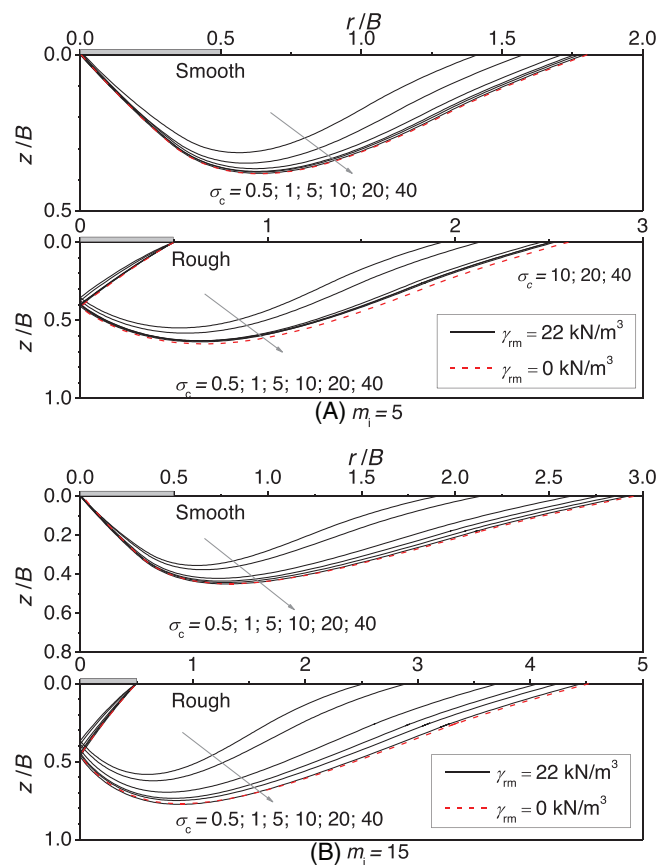


FIGURE 9 Effect of σ_c on size of failure surface under circular foundation: (A) $m_i = 5$; and (B) $m_i = 15$.

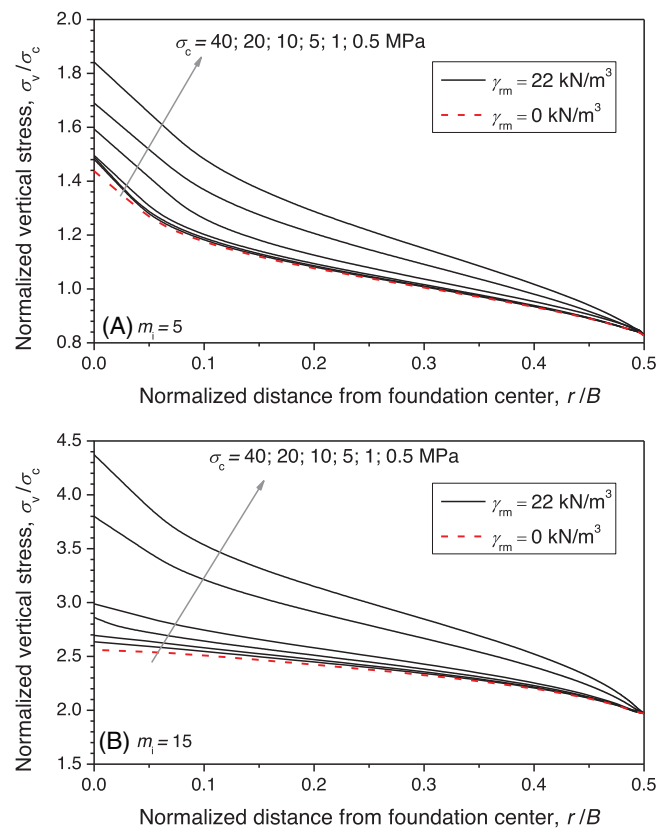


FIGURE 10 Effect of σ_c on vertical stress distribution on a smooth foundation base: (A) $m_i = 5$; and (B) $m_i = 15$.

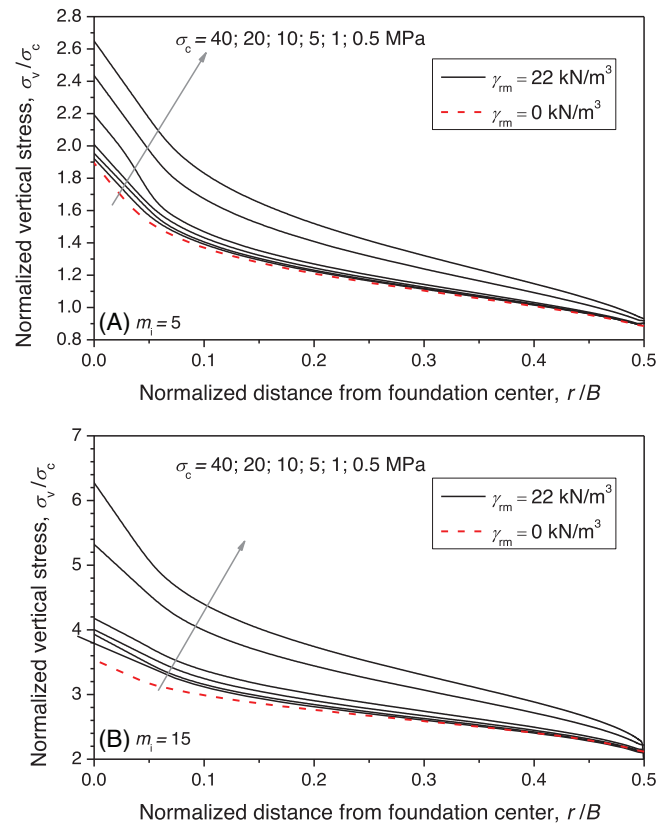


FIGURE 11 Effect of σ_c on stress distribution on a rough foundation base: (A) $m_i = 5$; and (B) $m_i = 15$.

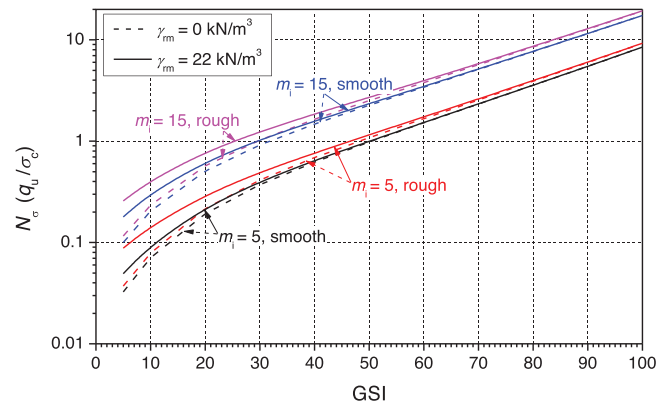


FIGURE 12 Relation between N_σ and GSI.

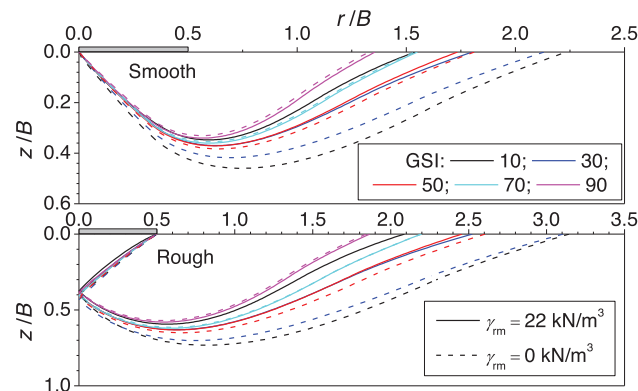


FIGURE 13 Effects of GSI on size of failure surface with $m_i = 5$.

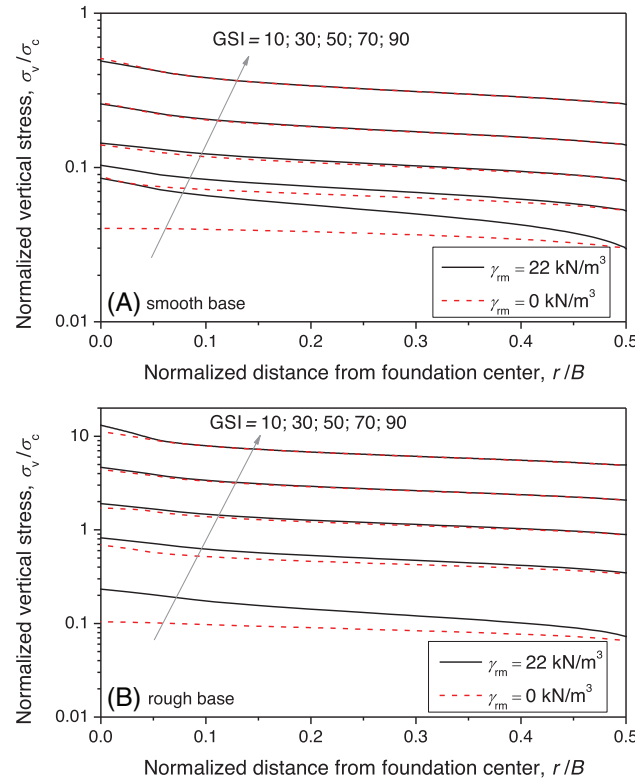


FIGURE 14 Effect of GSI on vertical stress distribution on foundation base with $m_i = 5$: (A) smooth base; and (B) rough base.

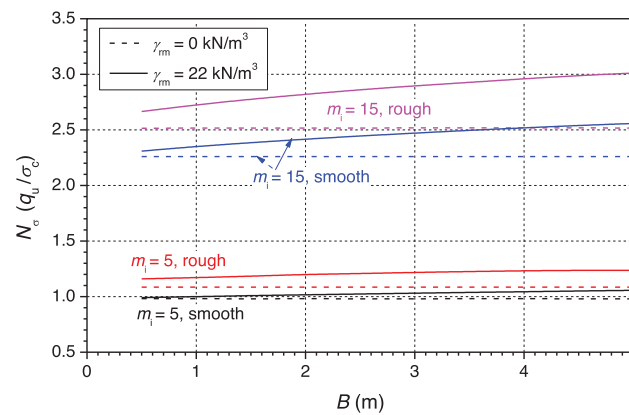


FIGURE 15 Relation between N_σ and B .

similar results can be found for $m_i = 15$. At both $\gamma_{rm} = 0$ and 22 kN/m^3 , and for both base conditions, the normalized vertical stress decreases with increasing distance from the center of the foundation. Generally, the normalized vertical stress at $\gamma_{rm} > 0$ is larger than that at $\gamma_{rm} = 0$, and the difference between the two conditions narrows with higher GSI because the effect of γ_{rm} decreases with the increase of GSI. Again, as expected, the normalized vertical stress at a rough base condition is larger than that at a smooth base condition with the same m_i , GSI and γ_{rm} .

6.3 | Effect of foundation size B

Figure 15 illustrates the effect of foundation diameter B on N_σ with $m_i = 5$ and 15 , $\gamma_{rm} = 0$ and 22 kN/m^3 , and both smooth and rough base conditions. The N_σ increases with a larger B when $\gamma_{rm} > 0$ but does not change with B if $\gamma_{rm} = 0$. At the same B , the N_σ increases with higher m_i , larger γ_{rm} and when the base condition changes from smooth to rough. It is thus important to consider these different factors in the design of a circular foundation on rock mass.

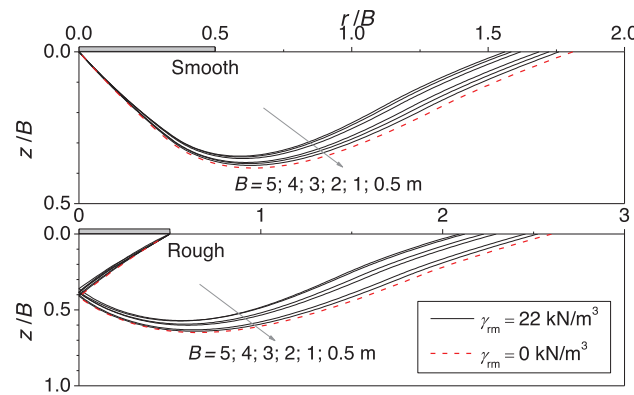


FIGURE 16 Effect of B on size of failure surface with $m_i = 5$.

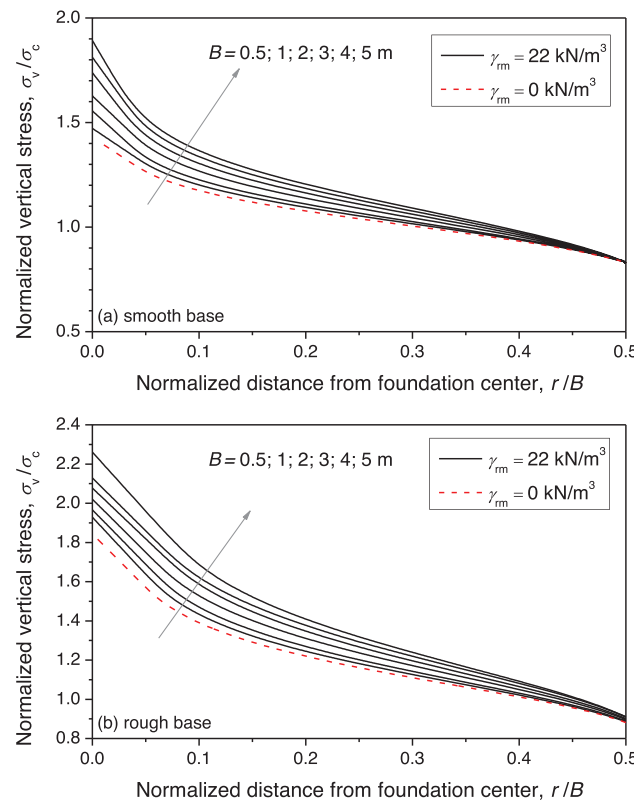


FIGURE 17 Effect of B on stress distribution on foundation base with $m_i = 5$: (A) smooth base and (B) rough base.

Figure 16 illustrates the effect of B on the normalized failure surface with $m_i = 5$, $\gamma_{rm} = 0$ and 22 kN/m^3 , and both smooth and rough base conditions. Only the cases with $m_i = 5$ are shown in order to save space and similar results can be found for $m_i = 15$. For $\gamma_{rm} > 0$, the normalized failure surface shrinks when B increases. For $\gamma_{rm} = 0$, the normalized failure surface stays the same when B changes and is larger than that for $\gamma_{rm} > 0$. As expected, at the same B , the normalized failure surface shrinks when the base condition changes from rough to smooth.

Figure 17 shows the effect of B on the normalized vertical stress distribution on the foundation base with $m_i = 5$, $\gamma_{rm} = 0$ and 22 kN/m^3 , and both smooth and rough base conditions. Similar trends can be found for $m_i = 15$. The normalized vertical stress increases with larger B at $\gamma_{rm} = 22 \text{ kN/m}^3$ but stays the same at $\gamma_{rm} = 0$. On the other hand, the normalized vertical stress increases when γ_{rm} changes from 0 to 22 kN/m^3 , and the base condition from smooth to rough. The variance of vertical stress distribution increases with larger γ_{rm} and B . This again demonstrates that the various factors interactively affect the performance of a circular foundation and should be properly considered.

7 | CONCLUSION

A semi-analytical solution is proposed for determining the ultimate bearing capacity q_u of circular shallow foundations on rock mass considering the strength dependence on three principal stresses or 3D strength, unit weight of rock mass, and both smooth and rough base conditions. The proposed solution is verified by using it to analyze thirteen test foundations. To evaluate the effect of the 3D strength, the presented solution is compared with a solution based on the 2D HB criterion. Finally, the proposed solution is used to conduct parametric studies on the effects of rock mass properties (γ_{rm} , m_i , σ_c , and GSI) and foundation dimension B on the ultimate bearing capacity, failure surface size, and vertical stress distribution on the base. The following major conclusions can be made:

1. The proposed solution considering the 3D strength and both smooth and rough base conditions makes reasonable predictions of the q_u of the collected thirteen test foundations on rock mass.
2. It is important to consider the base condition in determining the q_u of a circular shallow foundation on rock mass. If no information is available, it is recommended to take the average of the smooth and rough base solutions for the analysis of a circular shallow foundation on rock mass.
3. It is important to consider the 3D strength and unit weight of rock mass in determining the q_u of a circular shallow foundation on rock mass. Ignoring the 3D strength or unit weight of rock mass could significantly underestimate the q_u .
4. The rock mass parameters (γ_{rm} , m_i , σ_c and GSI) and foundation dimension B are all important factors that affect the q_u , the failure surface size and the vertical stress distribution on the foundation base, and therefore should be properly considered in the design of circular shallow foundations on rock mass.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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